NNETE THAPEDI TEISE

Electrical Engineering Student
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Portfolio: https://nneteteise.github.io/

PROFESSIONAL PROFILE

A highly analytical, resilient Electrical Engineering student from Kuruman, Northern Cape, currently completing a Diploma in Engineering Technology in Electrical Engineering at CPUT. Demonstrates a verified upward academic trajectory, with recent distinctions in Software Development 2 (80%) and Engineering Skills 1 (75%). Brings strong hands-on experience in industrial automation, process control systems, and power grid fault simulations, and is committed to articulating into the Bachelor of Engineering Technology (BEngTech) in Electrical Engineering programme in 2027 to deepen engineering competencies within local heavy industrial and mining sectors.

TECHNICAL SKILLS & SOFTWARE PROFICIENCIES

- **Automation & PLC Programming:** Siemens TIA Portal V16 (latching, interlocking gate operations, counter modules S_CU/S_CD/S_CUD, Timers: S_Pulse, S_ODT, S_ODTS, S_PEXT, and S_OFFDT)
- **Engineering Simulation & Control:** MATLAB, Simulink, NI Multisim, Proteus, SimulIDE, TinkerCAD
- **Programming Languages:** Embedded C (Arduino circuits, serial communications via Wokwi), Dev C++
- **Design & Drawing:** AutoCAD (foundational layout and drafting)
- **Digital Logic & Circuit Design:** Logic gate IC circuits, 7-segment display driver design, TinkerCAD simulation to hardware build
- **PCB & Product Design:** PCB layout design (EasyEDA), PCB drilling and hand-soldering, 3D enclosure design (TinkerCAD)

KEY ENGINEERING PROJECTS

Power Systems Fault Analysis Optimisation | Power Systems 2

- In my graduate attributes (GA) assignment I had to design and simulate a 3-phase circuit consisting of a 3-phase power supply, transformer, protection relay, breaker and a fault to analyse and optimise 3-phase current faults (line-to-line, line-to-ground, and line-to-line-to-line faults).
- Conducted AC current measurement practicals, calculating input/output voltage, current, and active power distribution.

Process Control & Automation Design | Process Control 2

- Engineered a hardware-in-the-loop temperature controller using Simulink and an Arduino Uno microcontroller.
- Programmed functional logic blocks, industrial sequential systems, and motor protection timers using Siemens TIA Portal V18.

Telemetry & Communication Modulations | Communication Systems 2 and Microprocessor 2

- Simulated signal behaviour across modulation profiles including AM, FM, PPM, PAM, PWM, and PCM using MATLAB.
- Built and debugged embedded C systems via Wokwi for microprocessors using digital input/output interfaces.

Applied Engineering Mathematics Integration | Engineering Maths 2

- Designed a comprehensive swimming pool plumbing system using calculus integration to optimise water pump sizing, water regulation loops, and thermal monitoring parameters.

Digital Electronics & Logic Design | Electronics 1

- Built a 7-segment digital display driven by logic gate operation IC circuits, first simulating the circuit in TinkerCAD before constructing the physical hardware.

PCB Assembly – Buzzer Circuit | Engineering Skills 1

- Drilled the PCB board and soldered all circuit connections to construct a functional buzzer unit, gaining hands-on PCB fabrication and hand-soldering skills.

Product Design & Development | Engineering Design 2

- Designed a 3D enclosure for the project using TinkerCAD and laid out a custom PCB board using EasyEDA.
- Drilled the designed PCB board and soldered all the components onto it and prepared a business proposal covering the full project, from concept through to a market-ready product.

EDUCATION & CREDENTIALS

Diploma in Engineering Technology in Electrical Engineering | Cape Peninsula University of Technology (CPUT) *— 2025 – Present (Final Year)*

- Academic benchmark: 100% pass rate in core Electrical streams.
- Notable results: Engineering Mathematics 2 (69%), Electrical Engineering 1 (70%), Software Development 2 — Pass with Distinction (80%), Engineering Skills 1 — Pass with Distinction (75%).
- Academic goal: scheduled to register for BEngTech in Electrical Engineering in 2027.

Academic Matric Upgrading | Access For Success, Upington *— 2022*

- Upgraded National Senior Certificate Physical Sciences and Core Mathematics results to meet advanced technological engineering entry requirements.

National Senior Certificate (Matric) | Kalahari High School, Kuruman *— 2021*

WORK EXPERIENCE

Volunteer – Filing Assistant | Department of COGHSTA, Kuruman *— 1 Week*

- Maintained strict administrative confidentiality while archiving, indexing, and cataloguing official provincial records.

References available on request

LANGUAGES & PERSONAL INTERESTS

- **Languages:** English, Afrikaans, Setswana, isiZulu (partial)
- **Interests:** Coding, debate & public speaking, repairing mechanical components and electrical devices, reading, writing poetry